

VNet-to-VNet VPN using Azure VPN Gateway

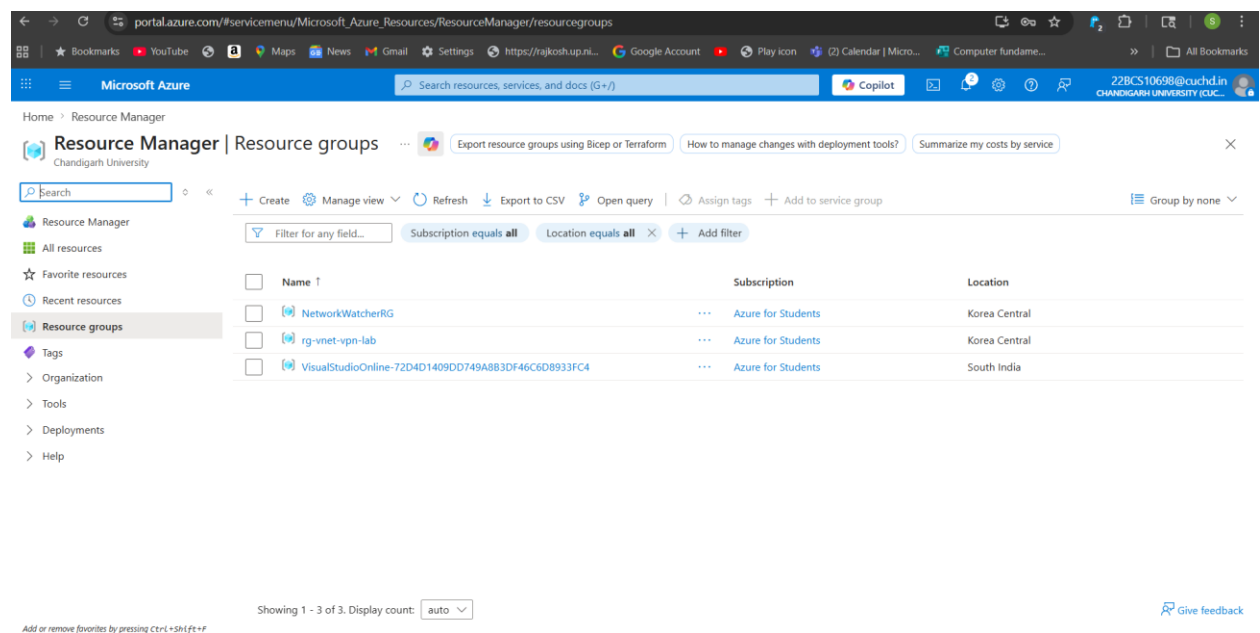
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Objective

To connect two Azure Virtual Networks securely using Azure VPN Gateway and verify communication between virtual machines using private IP addresses.

Step 1: Create Resource Group

Created resource group rg-vnet-vpn-lab in Azure.



The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and a user profile. The main content area displays the 'Resource Manager | Resource groups' page. A sidebar on the left lists navigation options: Resource Manager, All resources, Favorite resources, Recent resources, Resource groups (selected), Tags, Organization, Tools, Deployments, and Help. The main area shows a table of resource groups with columns for Name, Subscription, and Location. The table lists three resource groups: NetworkWatcherRG, rg-vnet-vpn-lab, and VisualStudioOnline-72D4D1409DD749A883DF46C6D8933FC4. The 'rg-vnet-vpn-lab' resource group is highlighted. Below the table, it indicates 'Showing 1 - 3 of 3. Display count: auto'.

Name	Subscription	Location
NetworkWatcherRG	Azure for Students	Korea Central
rg-vnet-vpn-lab	Azure for Students	Korea Central
VisualStudioOnline-72D4D1409DD749A883DF46C6D8933FC4	Azure for Students	South India

Step 2: Create Virtual Networks

Created vnet1 (10.0.0.0/16) and vnet2 (10.1.0.0/16).

Home > Network foundation

Network foundation | Virtual networks

Preview

Search

+ Create Manage view Refresh Export to CSV Open query Assign tags Add to service group Group by none

Filter for any field... Subscription equals all Resource Group equals all Location equals all Add filter

Name ↑	Resource Group	Location	Subscription
vnet1	rg-vnet-vpn-lab	Korea Central	Azure for Students
vnet2	NetworkWatcherRG	Korea Central	Azure for Students

Showing 1 - 2 of 2. Display count: auto

Add or remove favorites by pressing Ctrl+Shift+F

Give feedback

Step 3: Configure Subnets

Added subnet1 and GatewaySubnet in each VNet.

vnet1 | Subnets

Virtual network

Search

+ Subnet Refresh Manage users Delete Export to CSV

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Search subnets

Name ↑	IPv4	IPv6	Available IPs	Delegated to	Se
subnet1	10.0.1.0/24	-	250	-	-
GatewaySu...	10.0.255.0/27	-	availability ...	-	-

Showing 2 subnets

Give feedback

Add or remove favorites by pressing Ctrl+Shift+F

Step 5: Create Connections

Configured VNet-to-VNet connections and verified Connected status.

The screenshot shows the Azure portal interface. On the left, the navigation pane is open, and the 'Connections' option under 'Settings' is selected. The main area displays a table of VNet-to-VNet connections. At the top of the table is a search bar labeled 'Search connections'. The table has columns for Name, Status, Connection type, and Peer. Two connections are listed: 'vnet2-to-vnet1' and 'vnet1-to-vnet2', both with a status of 'Connected' and a connection type of 'VNet-to-VNet'. The peer for both is 'vnet2-gateway'. Below the table, there is a note: 'Add or remove favorites by pressing Ctrl+Shift+F'.

Name	Status	Connection type	Peer
vnet2-to-vnet1	Connected	VNet-to-VNet	vnet2-gateway
vnet1-to-vnet2	Connected	VNet-to-VNet	vnet2-gateway

Step 6: Create Virtual Machines

Created vm1 and vm2.

The screenshot shows the Azure portal interface for 'Virtual machines'. The left navigation pane is open, and 'Virtual machines' is selected. The main area displays a table of virtual machines. At the top of the table is a search bar labeled 'Search'. The table has columns for Name, Subscription, Resource Group, Location, Status, Operating system, Size, Public IP address, and Disks. Two virtual machines are listed: 'vm1' and 'vm2', both with a status of 'Running' and an operating system of 'Linux'. The size for both is 'Standard_B2ats...'. The public IP address for 'vm1' is '20.196.72.213' and for 'vm2' is '-'. The disks for both are '1'. Below the table, there is a note: 'Add or remove favorites by pressing Ctrl+Shift+F'.

Name	Subscription	Resource Group	Location	Status	Operating system	Size	Public IP address	Disks
vm1	Azure for Stude...	rg-vnet-vpn-lab	Korea Central	Running	Linux	Standard_B2ats...	20.196.72.213	1
vm2	Azure for Stude...	rg-vnet-vpn-lab	Korea Central	Running	Linux	Standard_B2ats...	-	1

Step 7: Test Connectivity

Verified ping between VMs using private IP addresses.

```
azureuser@vm1:~$ ping 10.1.1.4
PING 10.1.1.4 (10.1.1.4) 56(84) bytes of data.
64 bytes from 10.1.1.4: icmp_seq=1 ttl=64 time=2.71 ms
64 bytes from 10.1.1.4: icmp_seq=2 ttl=64 time=2.55 ms
64 bytes from 10.1.1.4: icmp_seq=3 ttl=64 time=2.06 ms
64 bytes from 10.1.1.4: icmp_seq=4 ttl=64 time=1.99 ms
64 bytes from 10.1.1.4: icmp_seq=5 ttl=64 time=2.40 ms
64 bytes from 10.1.1.4: icmp_seq=6 ttl=64 time=2.10 ms
64 bytes from 10.1.1.4: icmp_seq=7 ttl=64 time=6.13 ms
64 bytes from 10.1.1.4: icmp_seq=8 ttl=64 time=2.13 ms
64 bytes from 10.1.1.4: icmp_seq=9 ttl=64 time=1.97 ms
64 bytes from 10.1.1.4: icmp_seq=10 ttl=64 time=2.08 ms
64 bytes from 10.1.1.4: icmp_seq=11 ttl=64 time=2.05 ms
64 bytes from 10.1.1.4: icmp_seq=12 ttl=64 time=2.05 ms
64 bytes from 10.1.1.4: icmp_seq=13 ttl=64 time=2.38 ms
64 bytes from 10.1.1.4: icmp_seq=14 ttl=64 time=2.08 ms
```

```
azureuser@vm2:~$ ping 10.1.1.4
PING 10.1.1.4 (10.1.1.4) 56(84) bytes of data.
64 bytes from 10.1.1.4: icmp_seq=1 ttl=64 time=0.022 ms
64 bytes from 10.1.1.4: icmp_seq=2 ttl=64 time=0.022 ms
64 bytes from 10.1.1.4: icmp_seq=3 ttl=64 time=0.018 ms
64 bytes from 10.1.1.4: icmp_seq=4 ttl=64 time=0.017 ms
64 bytes from 10.1.1.4: icmp_seq=5 ttl=64 time=0.022 ms
64 bytes from 10.1.1.4: icmp_seq=6 ttl=64 time=0.019 ms
64 bytes from 10.1.1.4: icmp_seq=7 ttl=64 time=0.021 ms
64 bytes from 10.1.1.4: icmp_seq=8 ttl=64 time=0.028 ms
64 bytes from 10.1.1.4: icmp_seq=9 ttl=64 time=0.016 ms
64 bytes from 10.1.1.4: icmp_seq=10 ttl=64 time=0.015 ms
64 bytes from 10.1.1.4: icmp_seq=11 ttl=64 time=0.022 ms
64 bytes from 10.1.1.4: icmp_seq=12 ttl=64 time=0.016 ms
^C
--- 10.1.1.4 ping statistics ---
12 packets transmitted, 12 received, 0% packet loss, time 11261ms
rtt min/avg/max/mdev = 0.015/0.019/0.028/0.003 ms
```

Result

The two Azure VNets were successfully connected through a secure VPN tunnel. Communication between vm1 and vm2 was verified using ping over private IP addresses.